

NATIONAL QUANTUM MISSION AND QUANTUM TECHNOLOGY

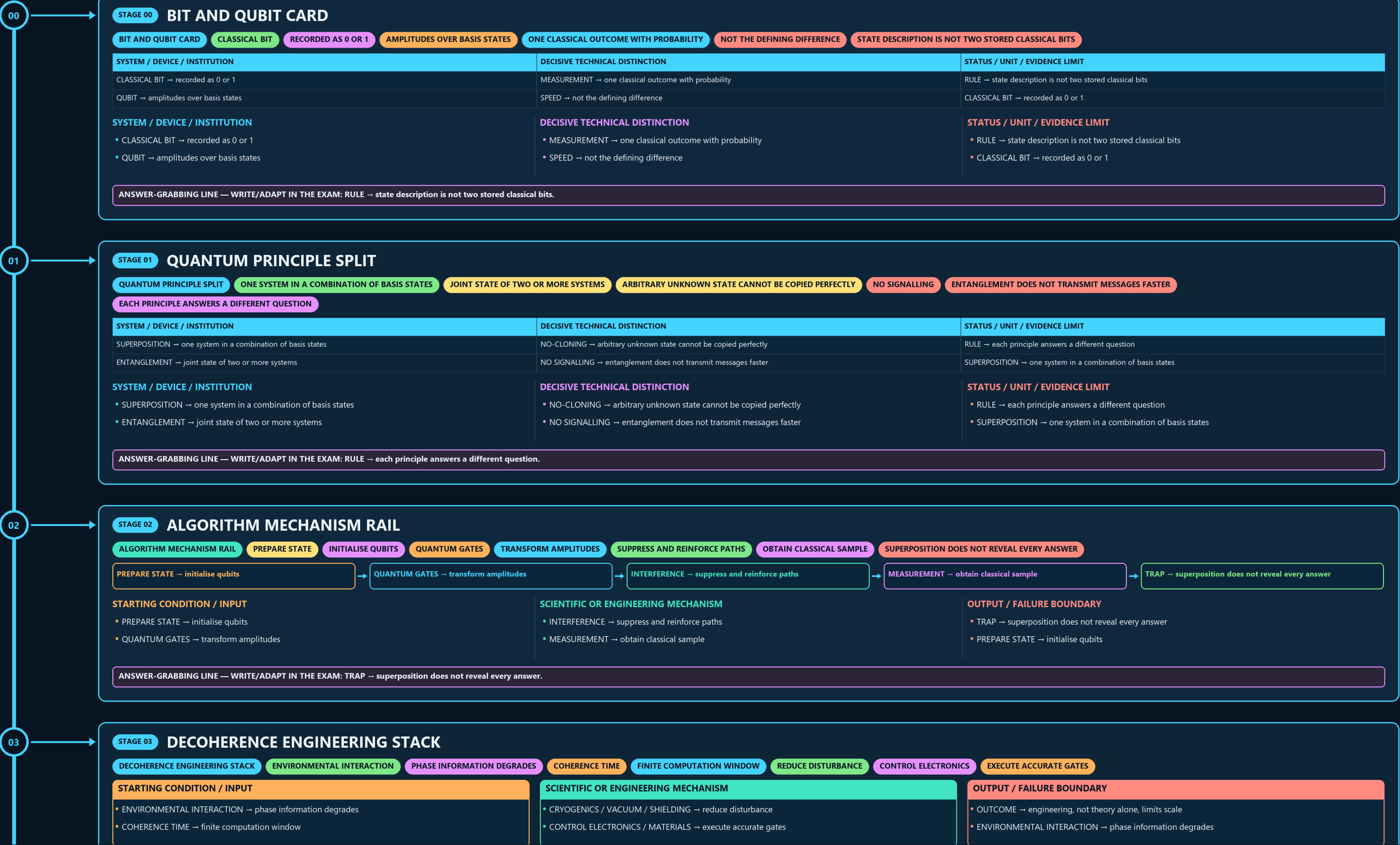
BIT AND QUBIT CARD → QUANTUM PRINCIPLE SPLIT → ALGORITHM MECHANISM RAIL → DECOHERENCE ENGINEERING STACK → PHYSICAL-TO-LOGICAL LADDER → FOUR TECHNOLOGY FAMILIES

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- PREPARE STATE → initialise qubits
- QUANTUM GATES → transform amplitudes

- INTERFERENCE → suppress and reinforce paths
- MEASUREMENT → obtain classical sample

- TRAP → superposition does not reveal every answer
- PREPARE STATE → initialise qubits

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → superposition does not reveal every answer.

03

STAGE 03

DECOHERENCE ENGINEERING STACK

ENVIRONMENTAL INTERACTION

PHASE INFORMATION DEGRADES

COHERENCE TIME

FINITE COMPUTATION WINDOW

REDUCE DISTURBANCE

CONTROL ELECTRONICS

EXECUTE ACCURATE GATES

STARTING CONDITION / INPUT

ENVIRONMENTAL INTERACTION → phase information degrades

COHERENCE TIME → finite computation window

SCIENTIFIC OR ENGINEERING MECHANISM

CRYOGENICS / VACUUM / SHIELDING → reduce disturbance

CONTROL ELECTRONICS / MATERIALS → execute accurate gates

OUTPUT / FAILURE BOUNDARY

OUTCOME → engineering, not theory alone, limits scale

ENVIRONMENTAL INTERACTION → phase information degrades

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → engineering, not theory alone, limits scale.

04

STAGE 04

PHYSICAL-TO-LOGICAL LADDER

PHYSICAL-TO-LOGICAL LADDER

PHYSICAL QUBIT

NOISY HARDWARE ELEMENT

ERROR CHARACTERISATION

MEASURE FAILURE CHANNELS

ERROR-CORRECTING CODE

REDUNDANT ENCODING

LOGICAL QUBIT

PHYSICAL QUBIT → noisy hardware element

ERROR CHARACTERISATION → measure failure channels

ERROR-CORRECTING CODE → redundant encoding

LOGICAL QUBIT → protected information unit

STARTING CONDITION / INPUT

PHYSICAL QUBIT → noisy hardware element

ERROR CHARACTERISATION → measure failure channels

SCIENTIFIC OR ENGINEERING MECHANISM

ERROR-CORRECTING CODE → redundant encoding

LOGICAL QUBIT → protected information unit

OUTPUT / FAILURE BOUNDARY

RULE → raw qubit count is not useful capability

PHYSICAL QUBIT → noisy hardware element

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → raw qubit count is not useful capability.

05

STAGE 05

FOUR TECHNOLOGY FAMILIES

FOUR TECHNOLOGY FAMILIES

SELECTED INFORMATION-PROCESSING ALGORITHMS

QUANTUM-STATE LINKS AND KEY DISTRIBUTION

PRECISION MEASUREMENT

ENABLING HARDWARE SUBSTRATE

QUANTUM TECHNOLOGY IS NOT COMPUTING ALONE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
COMPUTING → selected information-processing algorithms	SENSING / METROLOGY → precision measurement	TRAP → quantum technology is not computing alone
COMMUNICATION → quantum-state links and key distribution	MATERIALS / DEVICES → enabling hardware substrate	COMPUTING → selected information-processing algorithms

SYSTEM / DEVICE / INSTITUTION

COMPUTING → selected information-processing algorithms

COMMUNICATION → quantum-state links and key distribution

DECISIVE TECHNICAL DISTINCTION

SENSING / METROLOGY → precision measurement

MATERIALS / DEVICES → enabling hardware substrate

STATUS / UNIT / EVIDENCE LIMIT

TRAP → quantum technology is not computing alone

COMPUTING → selected information-processing algorithms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → quantum technology is not computing alone.

06

STAGE 06

QKD SECURITY CHAIN

QKD SECURITY CHAIN

QUANTUM STATES

DISTRIBUTE KEY MATERIAL

DISTURBANCE TEST

REVEAL POSSIBLE INTERCEPTION

AUTHENTICATED ENDPOINTS

PREVENT IMPERSONATION

CLASSICAL ENCRYPTION

QUANTUM STATES → distribute key material

DISTURBANCE TEST → reveal possible interception

AUTHENTICATED ENDPOINTS → prevent impersonation

CLASSICAL ENCRYPTION → protects message content

RULE → QKD alone is not a complete secure network

STARTING CONDITION / INPUT

QUANTUM STATES → distribute key material

DISTURBANCE TEST → reveal possible interception

SCIENTIFIC OR ENGINEERING MECHANISM

AUTHENTICATED ENDPOINTS → prevent impersonation

CLASSICAL ENCRYPTION → protects message content

OUTPUT / FAILURE BOUNDARY

RULE → QKD alone is not a complete secure network

QUANTUM STATES → distribute key material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.

06

STAGE 06

QKD SECURITY CHAIN

QKD SECURITY CHAIN

QUANTUM STATES

DISTRIBUTE KEY MATERIAL

DISTURBANCE TEST

REVEAL POSSIBLE INTERCEPTION

AUTHENTICATED ENDPOINTS

PREVENT IMPERSONATION

CLASSICAL ENCRYPTION

QUANTUM STATES → distribute key material

DISTURBANCE TEST → reveal possible interception

AUTHENTICATED ENDPOINTS → prevent impersonation

CLASSICAL ENCRYPTION → protects message content

RULE → QKD alone is not a complete secure network

STARTING CONDITION / INPUT

- QUANTUM STATES → distribute key material
- DISTURBANCE TEST → reveal possible interception

SCIENTIFIC OR ENGINEERING MECHANISM

- AUTHENTICATED ENDPOINTS → prevent impersonation
- CLASSICAL ENCRYPTION → protects message content

OUTPUT / FAILURE BOUNDARY

- RULE → QKD alone is not a complete secure network
- QUANTUM STATES → distribute key material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.

07

STAGE 07

QKD VERSUS PQC

QKD VERSUS PQC

QUANTUM HARDWARE AND PHYSICAL LINK

CLASSICAL QUANTUM-RESISTANT ALGORITHMS

QKD PURPOSE

SPECIALISED KEY DISTRIBUTION

PQC PURPOSE

BROAD STANDARDS AND SOFTWARE MIGRATION

COMPLEMENTARY TOOLS WITH DIFFERENT TIMELINES

SYSTEM / DEVICE / INSTITUTION

QKD → quantum hardware and physical link
PQC → classical quantum-resistant algorithms

DECISIVE TECHNICAL DISTINCTION

QKD PURPOSE → specialised key distribution
PQC PURPOSE → broad standards and software migration

STATUS / UNIT / EVIDENCE LIMIT

POLICY → complementary tools with different timelines
QKD → quantum hardware and physical link

SYSTEM / DEVICE / INSTITUTION

- QKD → quantum hardware and physical link
- PQC → classical quantum-resistant algorithms

DECISIVE TECHNICAL DISTINCTION

- QKD PURPOSE → specialised key distribution
- PQC PURPOSE → broad standards and software migration

STATUS / UNIT / EVIDENCE LIMIT

- POLICY → complementary tools with different timelines
- QKD → quantum hardware and physical link

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PQC PURPOSE → broad standards and software migration.

08

STAGE 08

NQM INSTITUTION MAP

NQM INSTITUTION MAP

MINISTRY OF SCIENCE AND TECHNOLOGY

MINISTERIAL UMBRELLA

IMPLEMENTING DEPARTMENT

NATIONAL CAPABILITY MISSION

FOUR VERTICAL ECOSYSTEM ANCHORS

DEPARTMENT, MISSION AND HUB ARE DISTINCT

MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

DST → implementing department

NQM → national capability mission

T-HUBS → four vertical ecosystem anchors

CONCEPT / ARCHITECTURE

- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella
- DST → implementing department

MECHANISM / APPLICATION

- NQM → national capability mission
- T-HUBS → four vertical ecosystem anchors

READINESS / STATUS LIMIT

- RULE → department, mission and hub are distinct
- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → department, mission and hub are distinct.

09

STAGE 09

FOUR T-HUB MAP

FOUR T-HUB MAP

IISc BENGALURU

QUANTUM COMPUTING

IIT MADRAS WITH C-DOT

QUANTUM COMMUNICATION

IIT BOMBAY

QUANTUM SENSING AND METROLOGY

IIT DELHI

SYSTEM / DEVICE / INSTITUTION

IISc BENGALURU → Quantum Computing
IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

IIT BOMBAY → Quantum Sensing and Metrology
IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

TRAP → hub creation is not operational capability
MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

SYSTEM / DEVICE / INSTITUTION

- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → hub creation is not operational capability
- MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.

10

STAGE 10

READINESS LADDER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
IISc BENGALURU → Quantum Computing	IIT BOMBAY → Quantum Sensing and Metrology	TRAP → hub creation is not operational capability
IIT MADRAS WITH C-DOT → Quantum Communication	IIT DELHI → Quantum Materials and Devices	MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

SYSTEM / DEVICE / INSTITUTION

- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → hub creation is not operational capability
- MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.

10

STAGE 10 **READINESS LADDER**

READINESS LADDER

CABINET APPROVAL

MISSION AUTHORITY AND OUTLAY

INTENDED FUTURE CAPABILITY

RESEARCH PROTOTYPE

BOUNDED DEMONSTRATION

VALIDATED SYSTEM

REPEATABLE PERFORMANCE EVIDENCE

CABINET APPROVAL → mission authority and outlay

TARGET → intended future capability

RESEARCH PROTOTYPE → bounded demonstration

VALIDATED SYSTEM → repeatable performance evidence

STARTING CONDITION / INPUT

- CABINET APPROVAL → mission authority and outlay
- TARGET → intended future capability

SCIENTIFIC OR ENGINEERING MECHANISM

- RESEARCH PROTOTYPE → bounded demonstration
- VALIDATED SYSTEM → repeatable performance evidence

OUTPUT / FAILURE BOUNDARY

- OPERATIONAL DEPLOYMENT → field use with sustainment
- CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful.

11

STAGE 11 **QUANTUM CLAIM FIREWALL**

QUANTUM CLAIM FIREWALL

QUBIT, SUPERPOSITION, ENTANGLEMENT AND DECOHERENCE

COMPUTING, COMMUNICATION, SENSING OR MATERIALS

PHYSICAL AND LOGICAL QUBITS; QKD AND PQC

APPROVAL, TARGET, PROTOTYPE, VALIDATION AND OPERATION

OUTLAY, QUBITS, DISTANCE AND STARTUP CLAIMS OFFICIALLY

EVIDENCE LIMIT

IDENTIFY PLATFORM, QUBIT/MEASUREMENT

DEFINE → qubit, superposition, entanglement and decoherence

CLASSIFY → computing, communication, sensing or materials

SEPARATE → physical and logical qubits; QKD and PQC

LABEL → approval, target, prototype, validation and operation

VERIFY → outlay, qubits, distance and startup claims officially

SYSTEM / DEVICE / INSTITUTION

- DEFINE → qubit, superposition, entanglement and decoherence
- CLASSIFY → computing, communication, sensing or materials

DECISIVE TECHNICAL DISTINCTION

- SEPARATE → physical and logical qubits; QKD and PQC
- LABEL → approval, target, prototype, validation and operation

STATUS / UNIT / EVIDENCE LIMIT

- VERIFY → outlay, qubits, distance and startup claims officially
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement.

E

EXTRA **SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE**

OPTIONAL ADVANCED ENRICHMENT

QUANTUM CAPABILITY IS PATH-DEPENDENT

NATIONS THAT BUILD TALENT, FABRICATION KNOW-HOW AND STANDARDS EARLY

THE MOST BINDING CONSTRAINTS ARE OFTEN ENGINEERING AND ECOSYSTEM

A MISSION APPROACH IS JUSTIFIED BECAUSE FRAGMENTED PROJECTS ALONE

NATIONAL MISSION FRAMEWORK.

T-HUB AT IISC BENGALURU

QUANTUM COMPUTING.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Quantum capability is path-dependent: nations that build talent, fabrication know-how and standards early can gain long-term advantages.
- CAUTION: The most binding constraints are often engineering and ecosystem constraints rather than pure theory.
- CAUTION: A mission approach is justified because fragmented projects alone may not create platform-level capability.
- NQM: national mission framework.

READINESS, UNIT OR STATUS LIMIT

- T-Hub at IISc Bengaluru: Quantum Computing.
- T-Hub at IIT Madras with C-DOT: Quantum Communication.
- T-Hub at IIT Bombay: Quantum Sensing & Metrology.
- T-Hub at IIT Delhi: Quantum Materials & Devices.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- PIB parliamentary answer: says the T-Hubs comprise 14 Technical Groups across 17 States and 2 UTs.
- Quantum communication is strategically relevant for secure networks and national-security communication contexts.
- Quantum sensing/metrology can matter for navigation, clocks, precision instrumentation and advanced measurement ecosystems.
- Quantum materials/devices are foundational because hardware dependence can otherwise cripple mission ambition.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.